

Improve Text Classification with Word Embedding — Group 22

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1 Introduction

This project investigates improving text classification performance using different word embedding techniques. A subset of the 2023 Amazon User Reviews and Item Metadata datasets was used, evaluating embeddings including Bag-of-Words (BoW), TF-IDF, Word2Vec, FastText, GloVe, and BERT. The study aims to determine which representations most effectively capture semantic meaning for sentiment classification tasks.

2 Exploratory Data Analysis: Data Inspection

2.1 Data Inspection Overview

The dataset consists of two primary components:

1. **User Review Data:** Individual user feedback including ratings and textual reviews.
2. **Item Metadata:** Product information such as title, price, and category.

Data originates from large `.jsonl` files for major categories (e.g., `All_Beauty`, `Amazon_Fashion`, `Video_Games`). A streaming approach was used to load a representative sample, extracting 20,000 rows for both review and metadata files across five categories, loaded into `unfiltered_df_reviews` and `unfiltered_df_meta`.

2.2 Features/Fields & Types

User Review Data

- Numerical: `rating` (float 1.0 to 5.0), `timestamp` (int), `helpful_vote` (int).
- Categorical: `title`, `text`, `asin`, `parent_asin`, `user_id`, `verified_purchase`.

Metadata

- Numerical: `average_rating`, `rating_number`, `price`.
- Categorical: `main_category`, `title`, `store`, `parent_asin`.
- List/Object: `features`, `description`, `images`, `videos`, `categories`, `details`, `bought_together`.

2.3 Missing Values and Basic Statistics

Several metadata fields (e.g., `features`, `description`, `videos`, `categories`, `subtitle`, `author`) contain missing values. Basic statistics like `average_rating` are pre-computed. The key numerical feature from review data is the `rating` column.

2.4 Data Preprocessing

Data cleaning and integration were performed in several steps:

2.4.1 Feature Selection

The first step involved reducing the dataset size by removing irrelevant or unnecessary columns from both the Review and Metadata DataFrames.

From the Review DataFrame (unfiltered_df_reviews):

- **Dropped:** `images`, `asin`, `user_id`, `timestamp`, `helpful_vote`
- **Reason:** These columns were removed to eliminate redundant identifiers (`asin`), high-cardinality user data (`user_id`), unnecessary media links (`images`), and data not required for the main analysis (`timestamp`, `helpful_vote`).

From the Metadata DataFrame (unfiltered_df_meta):

- **Dropped:** `bought_together`, `subtitle`, `author`, `videos`, `images`, `store`, `rating_number`, `categories`, `details`, `price`
- **Reason:** These fields were dropped because they were sparse (many missing values), contained complex nested data (`details`, `categories`), or were not relevant to the analysis.

2.4.2 Feature Engineering (Renaming)

To prevent naming conflicts during merging and improve clarity, columns with overlapping names were renamed as follows:

- `product_review_title` ← Review title
- `review` ← Review text
- `product_title` ← Metadata title

This renaming ensures that the merged dataset clearly distinguishes between the product’s descriptive information and the user-generated review content.

2.4.3 Data Integration (Merging)

The two cleaned DataFrames (`df_raw_review_filtered` and `df_meta_review_filtered`) were merged into a single DataFrame (`df`).

- **Method:** An inner join was performed using `pd.merge()`.
- **Key:** The `parent_asin` column was used as the common key, as it links reviews to their corresponding parent products.
- **Outcome:** This join created a unified dataset, ensuring that only reviews with matching product metadata were retained.

2.4.4 Handling Missing and Duplicate Data

1. Missing values handled with inner join and `df.dropna()`.
2. Duplicate rows removed using `df.loc[df.astype(str).drop_duplicates().index]`.

2.4.5 Standardization/Normalization

No standardization was performed in this project, as there was no need to apply any scaling that would improve the analysis

3 Visual Analysis

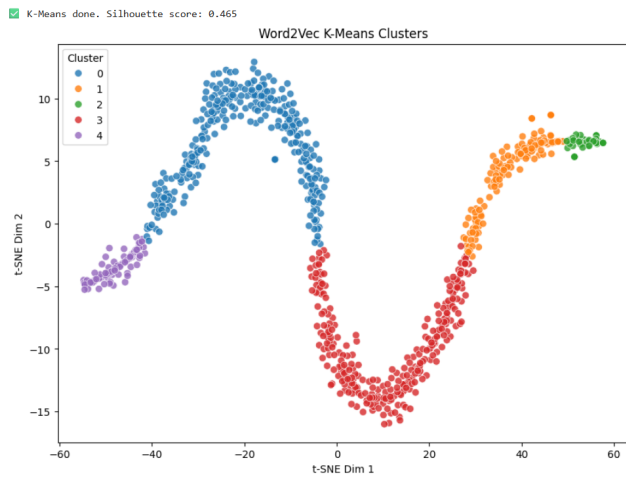


Figure 1: K-Means clustering of Word2Vec embeddings visualised with t-SNE, with distinct separation between semantic groups of words. The Word2Vec effectively captures meaningful relationships, resulting in richer representations for text classification.

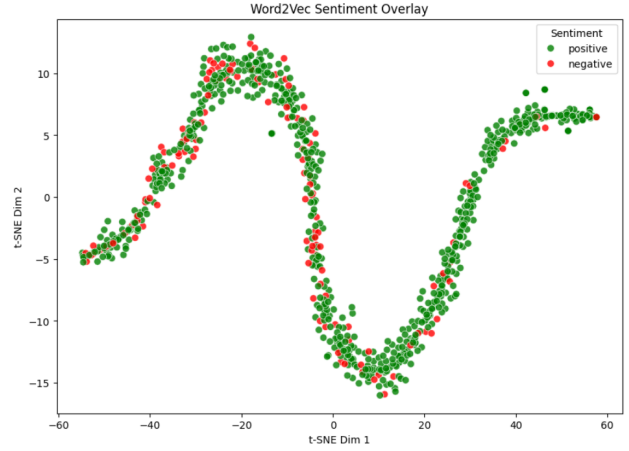


Figure 2: A clustering pattern with positive (green) and negative (red) sentiment distributions in the Word2Vec embedding space using t-SNE. The pattern shows how effectively Word2Vec captures the semantic and emotional relationships, with similar sentiments appearing closer together despite some expected overlap.

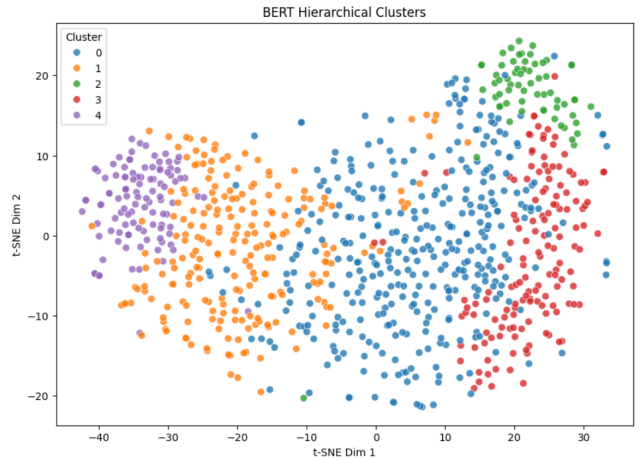


Figure 3: A hierarchical clustering of BERT sentence embeddings visualised with t-SNE, where each colour represents one of five clusters formed using Ward linkage. In comparison to Word2Vec, the BERT embeddings produce more distinct clusters reflecting a stronger contextual understanding and improved semantic separation that further enhances text classification performance.

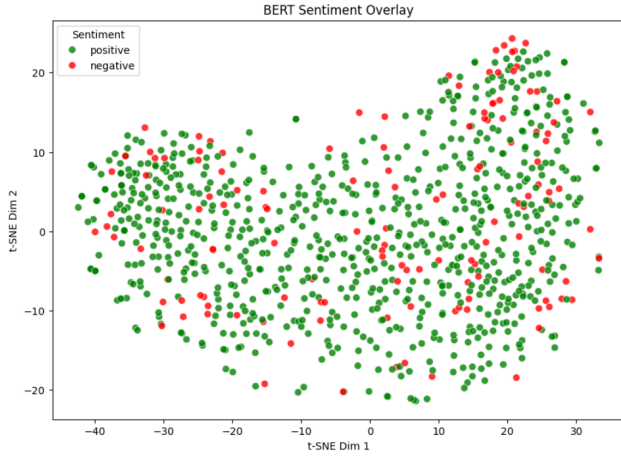


Figure 4: A t-SNE visualisation of BERT embeddings representing text samples by each point coloured by sentiment (green for positive, red for negative). The distribution reveals an imbalance and partial clustering by sentiment, indicating that BERT captures some sentiment structure but also broader semantic relationships, suggesting the need for more complex classifiers for optimal separation.

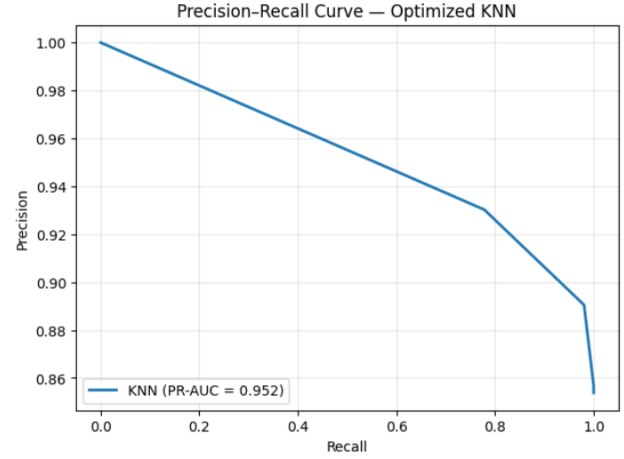


Figure 6: Precision-Recall curve of the optimized KNN model (PR-AUC = 0.953), indicating strong precision-recall balance.

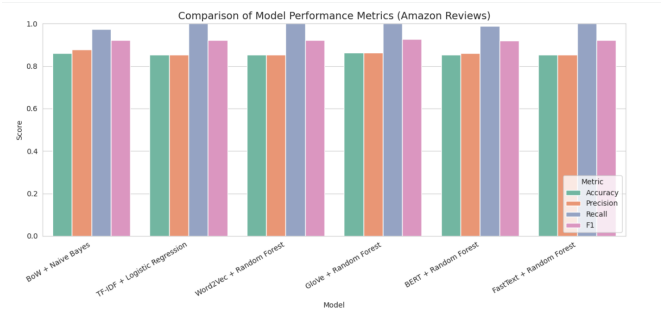


Figure 7: Comparison of classification performance across different word embedding approaches, showing accuracy, precision, recall, and F1-score

4 Results and Discussion

Table 1: Model Comparison Metrics

Model	Accuracy	Precision	Recall	F1
BoW + Naive Bayes	0.864	0.880	0.973	0.924
TF-IDF + Logistic Regression	0.854	0.854	1.000	0.921
Word2Vec + Random Forest	0.850	0.853	0.996	0.919
GloVe + Random Forest	0.864	0.862	1.000	0.926
BERT + Random Forest	0.854	0.861	0.988	0.920
FastText + Random Forest	0.847	0.853	0.992	0.917

Insights:

- **Class Imbalance Effect:** All models show high accuracy and strong performance for the majority positive class. However, the minority negative class is consistently under-predicted, resulting in low recall and F1-scores for negative reviews, particularly for TF-IDF, Word2Vec, and FastText models.

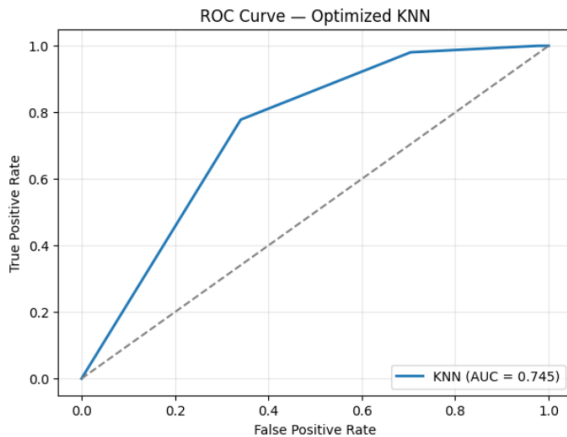


Figure 5: ROC curve of the optimized KNN model (AUC = 0.75), showing moderate class-separation performance.

- **Traditional embeddings (BoW, TF-IDF):** BoW + Naive Bayes performs reasonably well overall, achieving a high weighted F1-score (0.924) due to strong performance on positive reviews, but struggles to detect negative reviews. TF-IDF + Logistic Regression also favors the positive class entirely, failing to predict negative reviews.
- **Semantic embeddings (Word2Vec, GloVe, FastText):** Embeddings that capture semantic relationships improve weighted performance. GloVe + Random Forest attains the highest weighted F1-score (0.926), showing that classical embeddings combined with ensemble classifiers can effectively handle overall classification, despite challenges with negative-class detection.
- **Contextual embeddings (BERT):** BERT + Random Forest balances precision and recall for positive reviews, though negative-class detection remains limited (F1 = 0.12 for negative class). This demonstrates BERT’s ability to capture contextual information but highlights sensitivity to imbalanced datasets.
- **Practical Implications:** While overall metrics suggest strong performance across models, fine-grained evaluation indicates that additional strategies (e.g., oversampling, class weighting, or data augmentation) are required to improve detection of underrepresented classes.

5 Conclusion

This study evaluated multiple word embedding strategies for text classification on Amazon review data. Key conclusions are:

- Traditional embeddings like BoW and TF-IDF provide solid baseline performance, particularly for the majority positive class, but fail to adequately capture minority-class patterns.
- Semantic embeddings (Word2Vec, GloVe, FastText) improve overall predictive capability by leveraging word-level meaning. GloVe + Random Forest achieved the best overall weighted F1-score (0.926), demonstrating the effectiveness of combining classical embeddings with ensemble methods.
- Contextual embeddings (BERT) capture nuanced semantic relationships and maintain competitive performance, yet negative-class prediction remains limited due to class imbalance.
- Future work should focus on addressing class imbalance through methods such as oversampling, class weighting, or synthetic data generation. Additionally, hyperparameter optimization and ensemble modeling could further enhance performance across both majority and minority classes.

Overall, the results indicate that while contextual embeddings offer sophisticated language understanding, classical embeddings paired with robust classifiers remain highly effective for practical text classification tasks, especially in imbalanced datasets.

article

Some text.

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